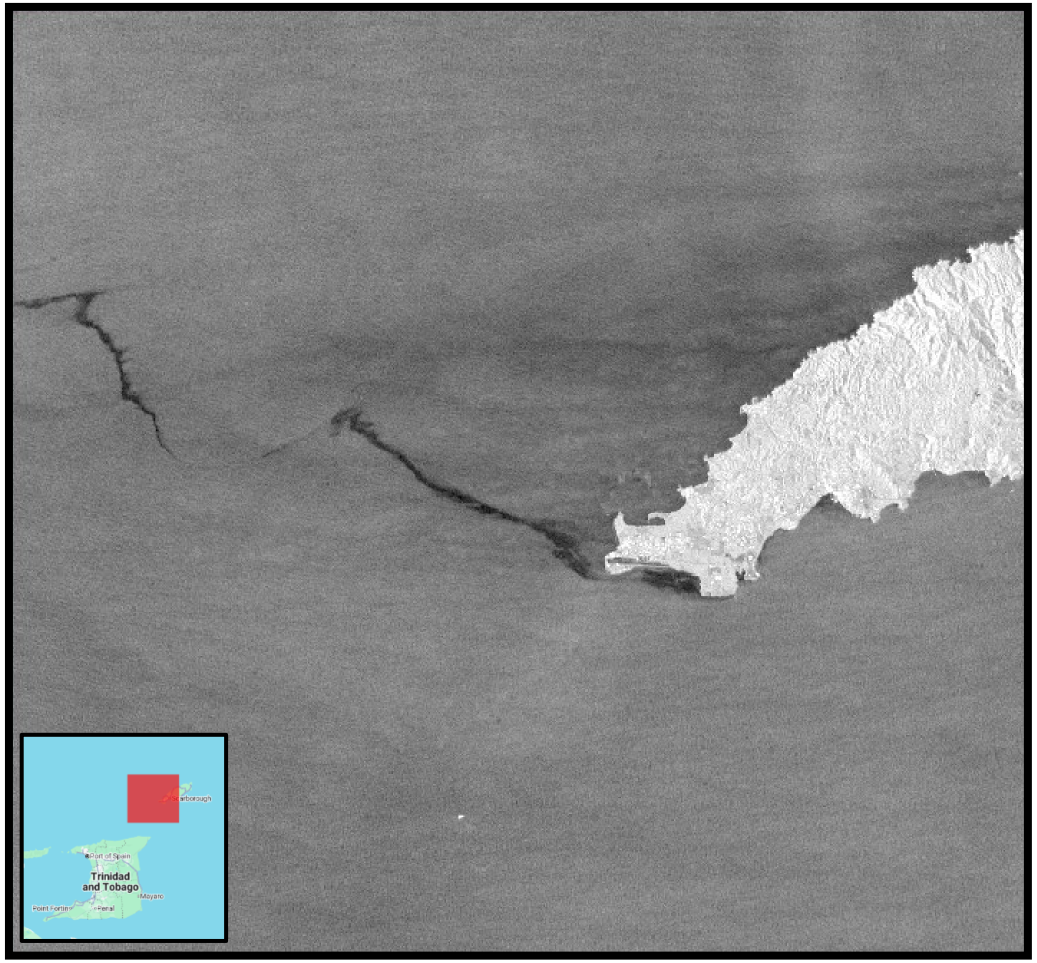
GIT 712: Assignment1 (GEE Time Series & Change Detection Analysis)



Time Series and change analysis of an oil spill on the coast of Tobago.

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Background of Ocean oil spills.

- Oil spills in the ocean are the result of accidents which involve oil tankers, pipelines and drilling rigs. As oil is less dense than water it lies on top of the water surface as seen in the figure on the right.

- This water contamination results in a few long-term consequences such as, poisoning of marine life, mangrove and vegetation loss, and disrupting marine ecosystems.

- Oil spills are dynamic as they move with wind and currents, this makes monitoring these events challenging.

- As urgency is needed for these devastating events, the usual method of boat-based surveys is slow. This is where remote sensing is needed to quantify and examine the effects of these oil spills.


Study Area

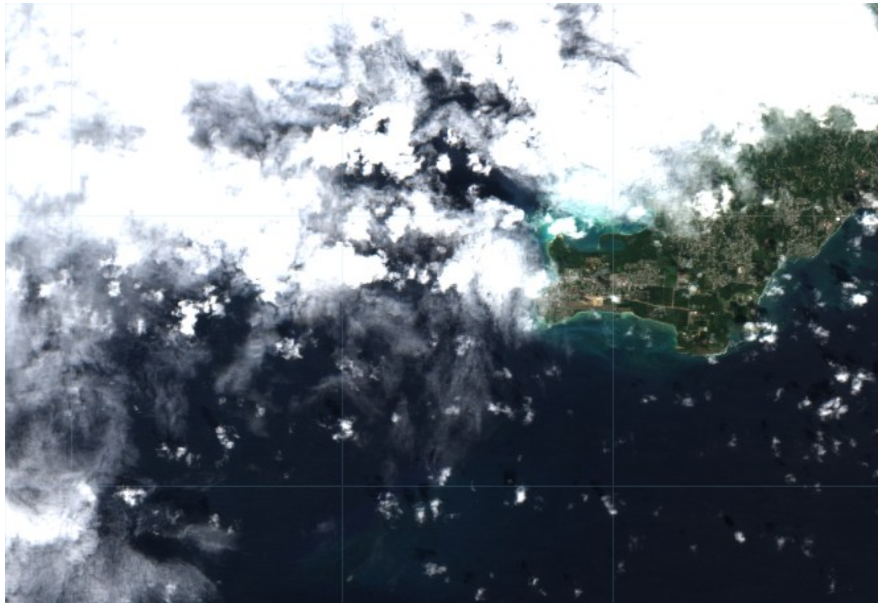
- This event occurred on the 7th of February 2024, where an overturned barge began leaking oil off the coast of the small island of Tobago.
- This oil spill ended up extending over 100km, the oil tanker was leaking for one month, but the environmental impact lasted much longer.
- The temporal window for this study starts in the beginning of January to April of 2024. This would provide data for the pre, during, and post oil spill.

Why SAR Data?

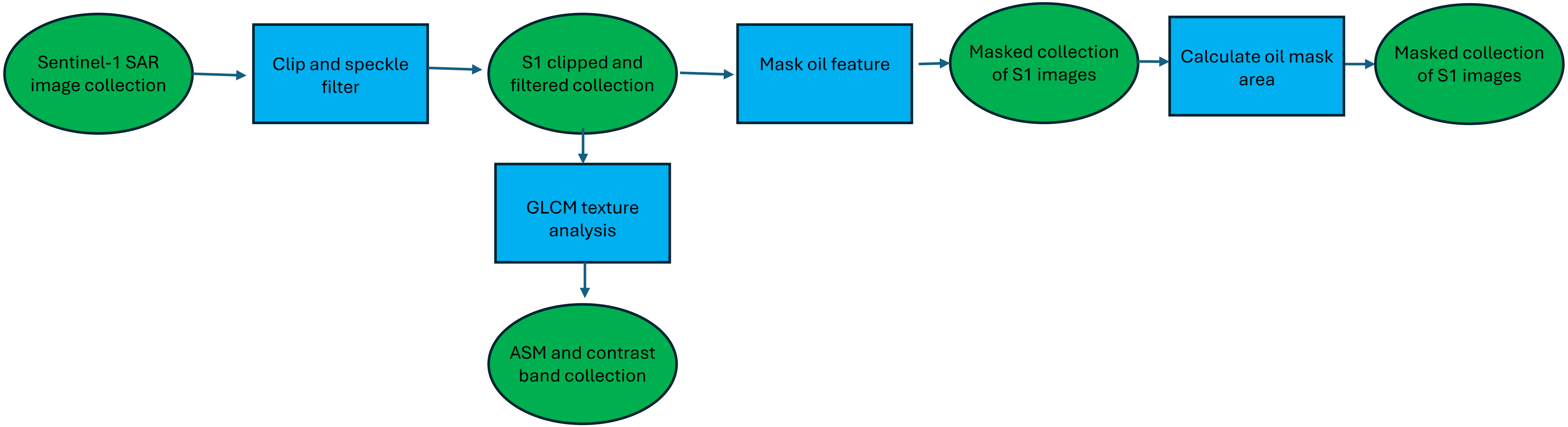
- The Sentinel-1 sensor is a dual polarized C-band sensor with a 5m x 20 m resolution.

- Unlike the Sentinel-2 optical sensor, the SAR sensor can see through the clouds. As the temporal window for detecting oil spills is short there is a need for fast and reliable imagery.

- SAR provides information of the physical properties of surfaces; this makes it sensitive to textures and surface roughness. This is a necessity in oil spill detection as it can differentiate between the smooth surface of the oil and the slightly rougher surface of the water

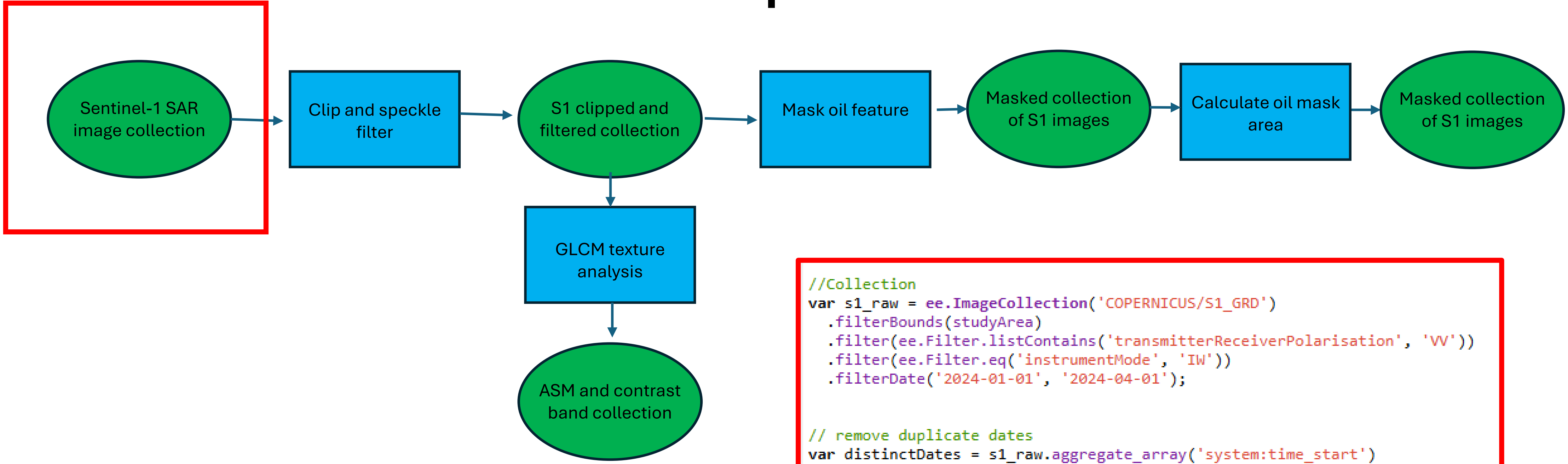


Methods



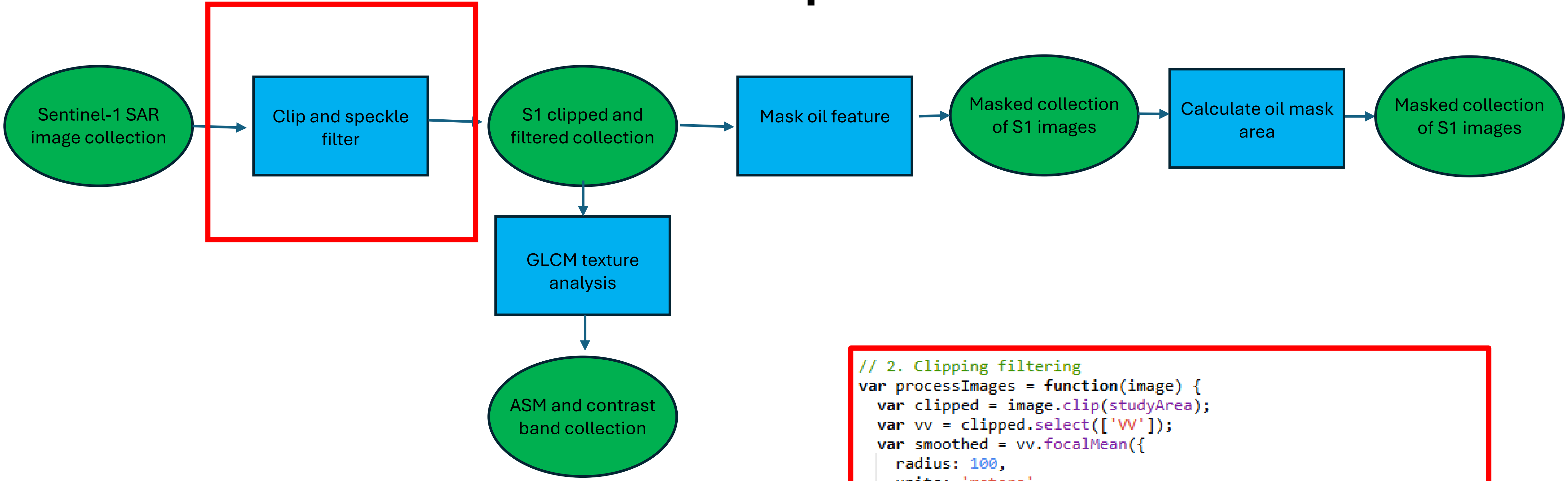
Sentinel-1 SAR image collection
Clip and speckle filter
S1 clipped and filtered collection
Mask oil feature
Masked collection of S1 images
Calculate oil mask area
Masked collection of S1 images
GLCM texture analysis
ASM and contrast band collection

Data-Preparation



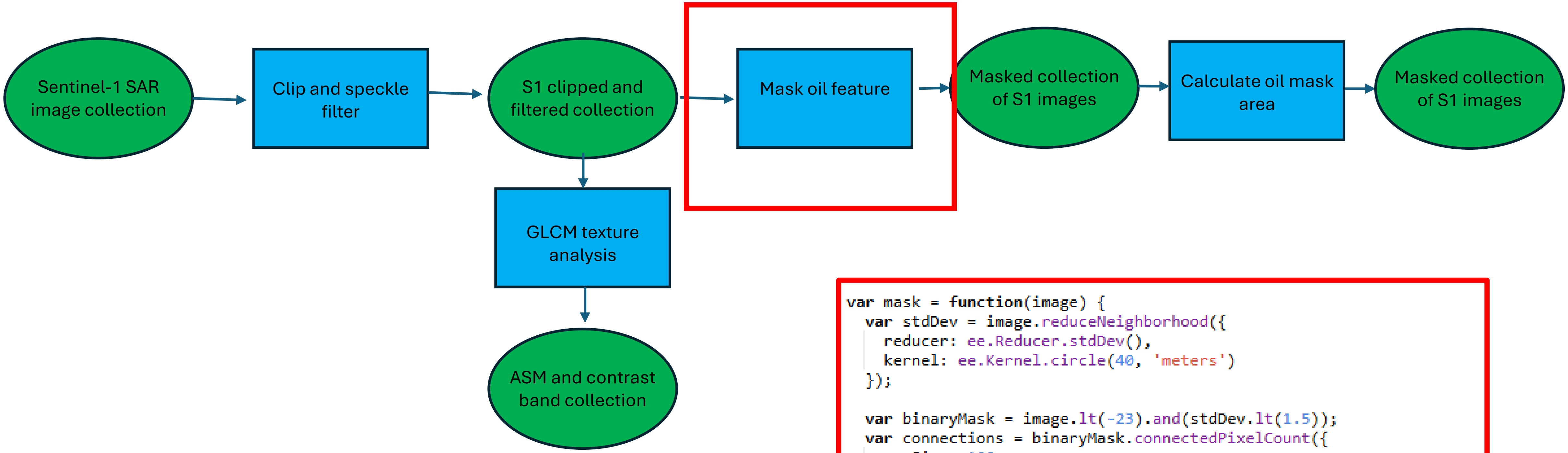

```
//Collection
var s1_raw = ee.ImageCollection('COPERNICUS/S1_GRD')
  .filterBounds(studyArea)
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
  .filter(ee.Filter.eq('instrumentMode', 'IW'))
  .filterDate('2024-01-01', '2024-04-01');


// remove duplicate dates
var distinctDates = s1_raw.aggregate_array('system:time_start')
  .map(function(d) { return ee.Date(d).format('YYYY-MM-dd') })
  .distinct();

// mosaicked image per day
var s1_collection = ee.ImageCollection(distinctDates.map(function(d) {
  var date = ee.Date(d);
  return s1_raw.filterDate(date, date.advance(1, 'day'))
    .mosaic()
    .set('system:time_start', date.millis())
    .set('date', d);
}));
```

Data-Preparation



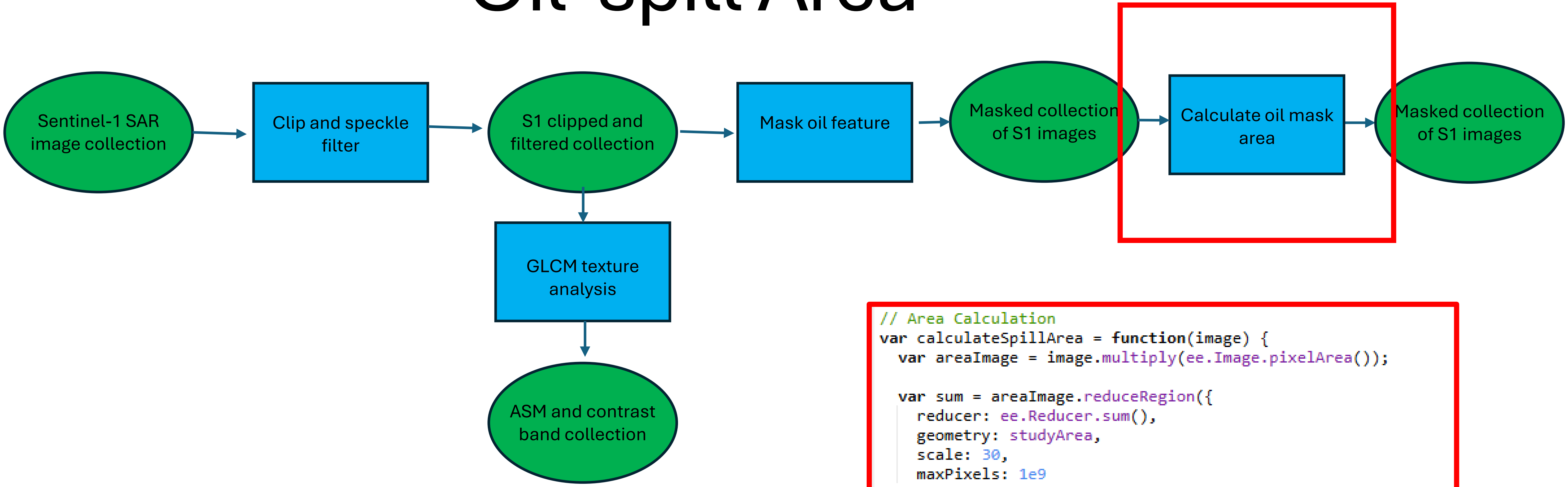

```
// 2. Clipping filtering
var processImages = function(image) {
  var clipped = image.clip(studyArea);
  var vv = clipped.select(['VV']);
  var smoothed = vv.focalMean({
    radius: 100,
    units: 'meters',
    kernelType: 'circle'
  });

  return smoothed.copyProperties(image, ['system:time_start']);
};

var collection_clip_filt = s1_collection.map(processImages);
```

Oil Masking



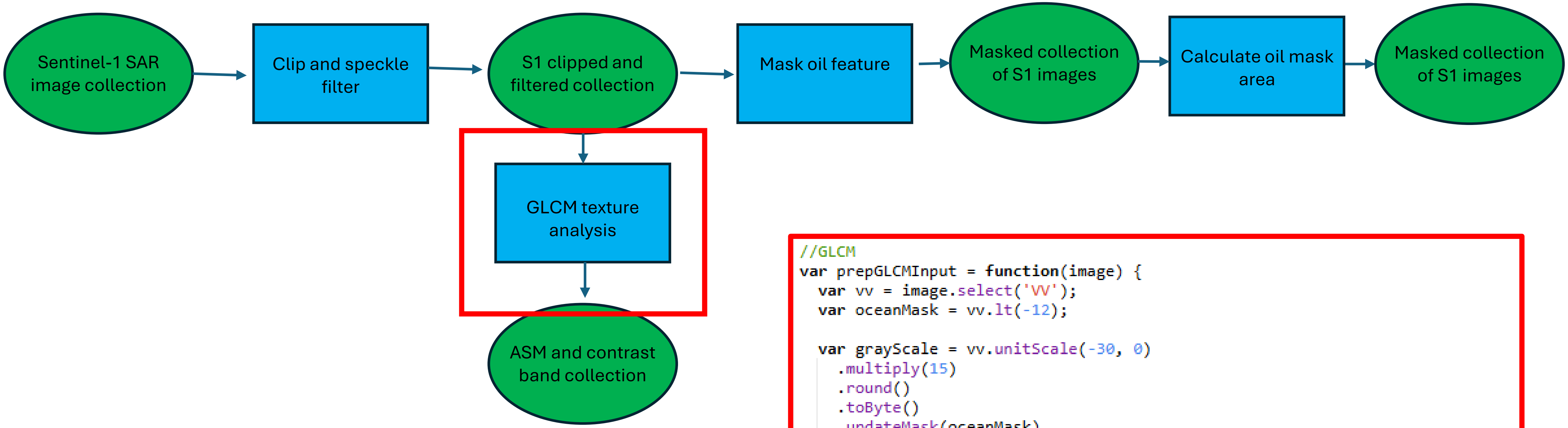

```javascript
var mask = function(image) {
  var stdDev = image.reduceNeighborhood({
    reducer: ee.Reducer.stdDev(),
    kernel: ee.Kernel.circle(40, 'meters')
  });

  var binaryMask = image.lt(-23).and(stdDev.lt(1.5));
  var connections = binaryMask.connectedPixelCount({
    maxSize: 100,
    eightConnected: true
  });

  var finalMask = binaryMask
    .updateMask(connections.gte(25))
    .unmask(0)
    .clip(studyArea)
    .rename('oil_mask_refined');


  return finalMask.copyProperties(image, ['system:time_start']);
};
```

Oil-spill Area

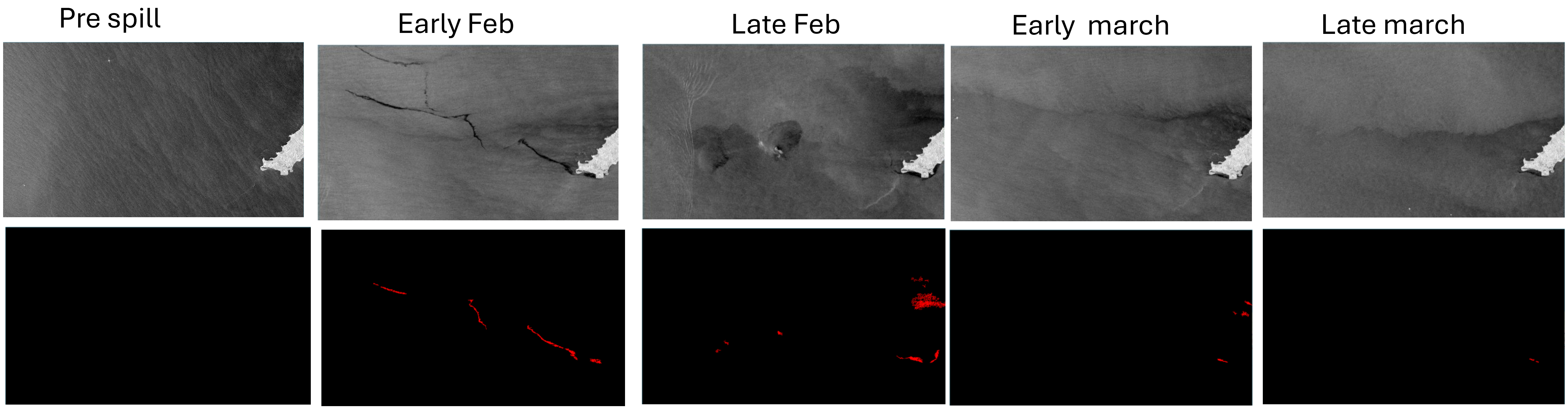


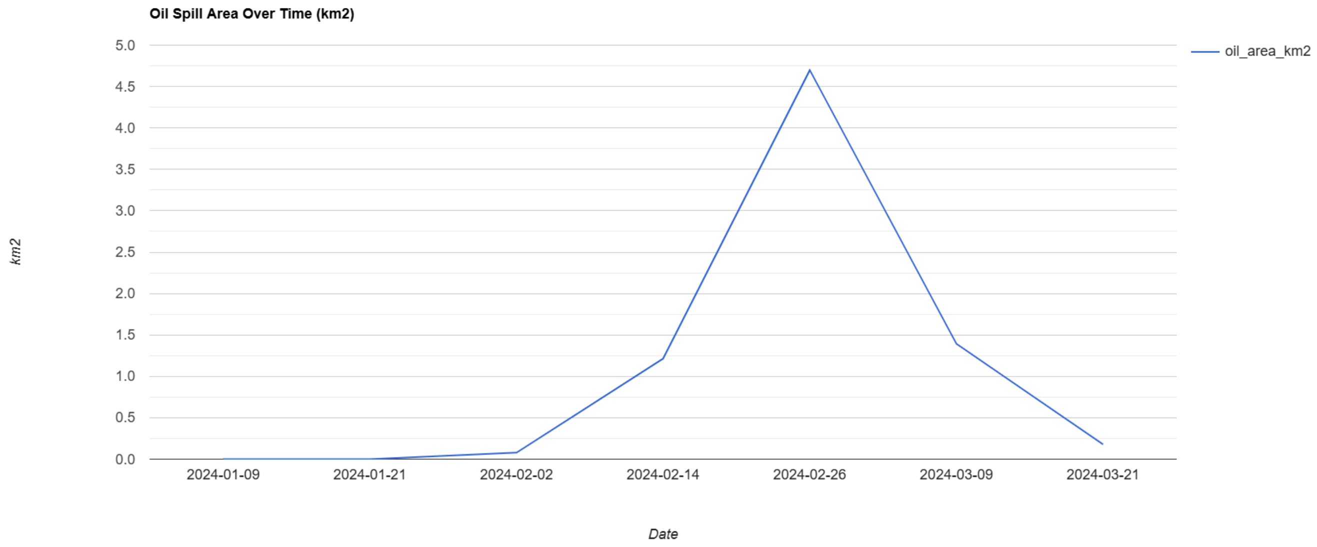

```
// Area Calculation
var calculateSpillArea = function(image) {
  var areaImage = image.multiply(ee.Image.pixelArea());

  var sum = areaImage.reduceRegion({
    reducer: ee.Reducer.sum(),
    geometry: studyArea,
    scale: 30,
    maxPixels: 1e9
  });

  var areaValue = ee.Number(sum.get('oil_mask_refined'));
  var areaKm2 = areaValue.divide(1e6);

  return image.set({
    'oil_area_km2': areaKm2,
    'date': image.date().format('YYYY-MM-dd')
  });
};

var col_Area = collection_mask.map(calculateSpillArea);
```

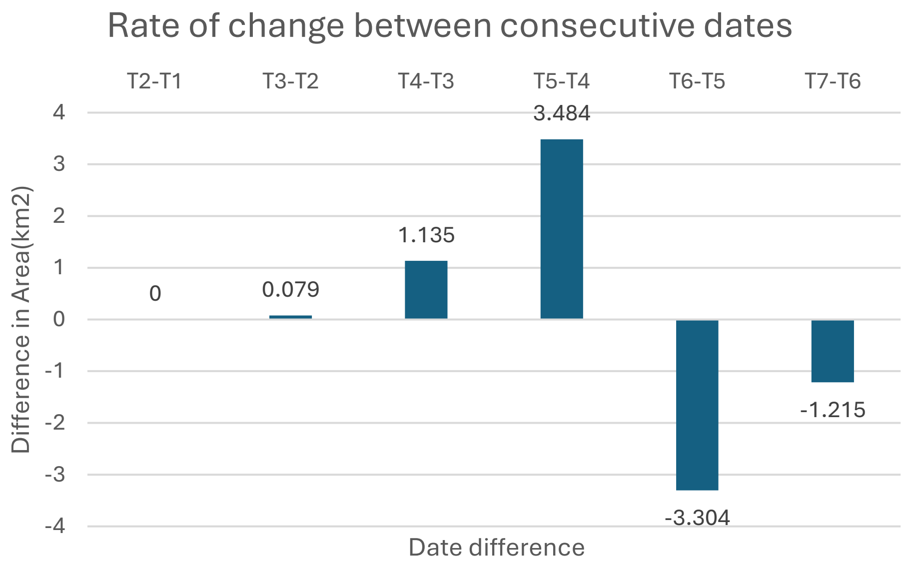
GLCM texture analysis



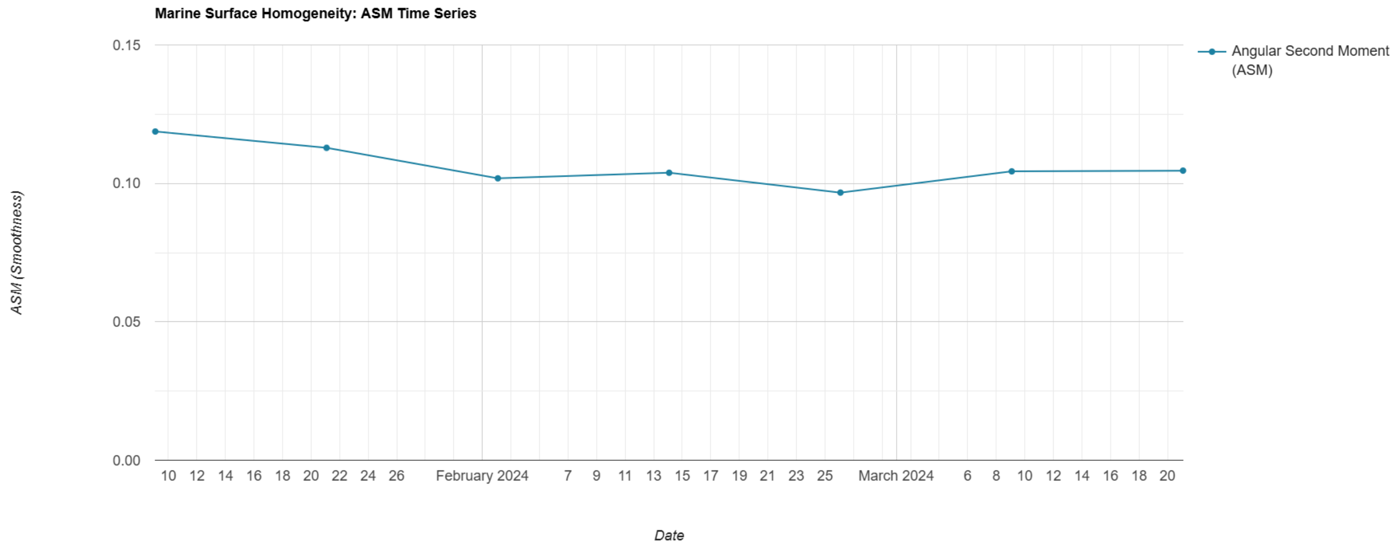

```
//GLCM
var prepGLCMInput = function(image) {
  var vv = image.select('VV');
  var oceanMask = vv.lt(-12);

  var grayScale = vv.unitScale(-30, 0)
    .multiply(15)
    .round()
    .toByte()
    .updateMask(oceanMask)
    .rename('VV_gray');

  return image.addBands(grayScale);
};

var addGLCM = function(image) {
  var glcm = image.select('VV_gray').glcmTexture({
    size: 4,
    average: true
  });
  return image.addBands(glcm);
};
var textureCol = collection_clip_filt.map(prepGLCMInput).map(addGLCM);
```

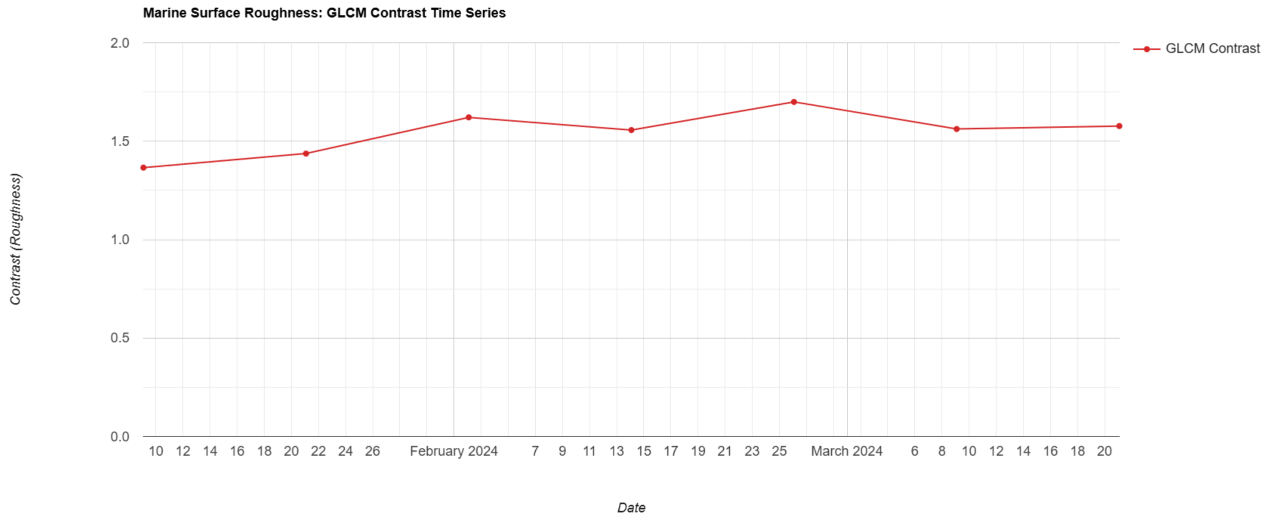

Pre spill
Early Feb
Late Feb
Early march
Late march


Oil Spill Area Over Time (km2)
oil_area_km2
km2
Date
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2024-01-09
2024-01-21
2024-02-02
2024-02-14
2024-02-26
2024-03-09
2024-03-21


Rate of change between consecutive dates
T2-T1
T3-T2
T4-T3
T5-T4
T6-T5
T7-T6
Difference in Area(km2)
Date difference
0
0.079
1.135
3.484
-3.304
-1.215
4
3
2
1
0
-1
-2
-3
-4

GLCM Results





Conclusion

- This study looked at the backscatter values as well as texture data to see how the oil spill moved over time. From this it is shown that over the time of the oil spill that the smoothness decreases and roughness increases and returns to the baseline.

- Within the study area the oil spill spread over 4.7 km^2. After this the SAR data shows that overtime it returns to similar values of the baseline.

- The use of GEE and SAR data to perform a time series analysis and change detection, allowed for the use of multiple months of data in a short span of time.

- The limitations with this study was the speckle nature of SAR imagery, this was seen in the low accuracy of masking .

- For future analysis I want to incorporate image fusion of Sentinel 1 and 2 data, this could improve the data temporally by having data at closer acquisition dates. This could also improve the classification of the oil spill.